IT24100327 - Perera B.P.N

Probability and Statistics – Lab 08

R 4.5.1 · C:/Users/hirum/OneDrive/Desktop/IT24100600/

```
> setwd("C:\\Users\\hirum\\OneDrive\\Desktop\\IT24100600")
> data <- read.table('Exercise - LaptopsWeights.txt', header=TRUE)
> names(data) <- c("weights")
> attach(data)
> popmn <- mean(weights)
> popsd <- sd(weights)
> samples <- c()
> n <- c()
> for (i in 1:25) {
+   s <- sample(weights, 6 , replace = TRUE)
+   samples <- cbind(samples, s)
+   n <- c(n, paste('S', i))
+ }
> colnames(samples) = n
> s.means <- apply(samples, 2, mean)
> s.sds <- apply(samples, 2, sd)
> samplemean <- mean(s.means)
> samplesd <- sd(s.means)
> popmn
[1] 2.468
> popsd
[1] 0.2561069
> samplemean
[1] 2.481467
> samplesd
[1] 0.07910424
> truesd <- popsd / sqrt(6)
> truesd
[1] 0.1045552
```